Ahnaf Adib

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Education

University of Texas at Dallas — Richardson, TX
Fast Track B.S./M.S. in Computer Science, GPA: 3.9/4.0 — Expected May 2028

Technical Skills

Languages: Python, C++, Java, SQL
Research: Diffusion Models, Mechanistic Interpretability, Knowledge Distillation, LoRA/PEFT, Multilingual NLP, Multi-Agent LLM Systems, Spatial/Bayesian ML
Frameworks & Systems: PyTorch, Hugging Face Transformers, NumPy/CuPy, CUDA, AWS, Modal, GCP, Kubernetes

Experience

Research Assistant — Richardson, TX
University of Texas at Dallas — *Feb 2025 – Present*
Advisor: Prof. Latifur Khan

- Built FirmVLA, a GPU serving system for heterogeneous VLA policies with an exact cyclic admission schedule; recorded zero SLO violations across six fleets (up to 122 robots) and raised task success under missed refills from 41.2% to 79.8%.
- Built XREAD, an interpretability-guided adapter placement for LLM knowledge distillation; matched a 7.9× larger LoRA recipe at 0.152% of trainable parameters across Qwen2, Llama-3.2, and Gemma-2.
- Pretrained 110M-parameter Spanish and Arabic ConfliBERT variants on 4× A100 GPUs; both reached 86% F1 on political-conflict and violence text classification.
- Benchmarked routed experts against attribution-selected neurons as intervention units across eight task families in Mixture-of-Experts models (Qwen3-30B-A3B, Gemma-4-26B-A4B, Mixtral-8x7B); expert suppression ran up to 9.5× larger than amplification.
- Built Quanta, a 9,032-example English-Arabic translation-error dataset with a 23-category taxonomy and an LLM-audited pipeline; trained an AraBERTv2 classifier to 61% F1 with 83–97% annotator agreement.
- Received $25K in NSA-sponsored research funding.

Machine Learning Researcher — Remote
Rutgers Global Health Institute — *Jan 2026 – Present*

- Led a nationwide Bayesian hierarchical spatial analysis of health vulnerability in 2,921 U.S. counties with Prof. Ubydul Haque; 3.4% carry severe burden across all 3 health domains at once.
- Built explainable Bayesian spatial models over 33 health outcomes and 41 county indicators; concentrated disadvantage was the strongest predictor, validated with spatial-block cross-validation.

Software Engineer Intern — Dallas, TX
Yobi — *Jun 2025 – Dec 2025*

- Deployed a production Python voice-agent system that handles 5,000 to 10,000 healthcare calls per week; escalation errors fell 60% and multi-intent latency dropped 300 ms.
- Increased successful AI tool execution from 70% to 85% through schema redesign, prompt iteration, and conversation-state tracking; cost per 1,000 calls fell 27%.
- Redesigned booking and follow-up workflows; completed patient bookings rose 22% and appointment no-shows fell 15%.
- Restructured multi-tenant infrastructure into tenant-isolated pods on GCP and Kubernetes; deployment time fell to under 5 minutes.

Summer Researcher — Remote
University of Illinois Urbana-Champaign — *Jun 2026 – Aug 2026*

- Benchmarked Dream-Coder, DiffuCoder, and LLaDA against autoregressive baselines on HumanEval, MBPP, and LiveCodeBench.
- Built adaptive remasking and backtracking for diffusion LLM decoding; cut decoding time 2 to 3× with no pass@1 loss, or about 5× with a 3-point pass@1 drop.

Selected Projects

VinTAGe: Video-and-Text-Conditioned Audio Generation (CVPR 2025) — sakshamsingh1/vintage_train_code

- Trained PyTorch pipelines that reduced FAD by 25% and FID by 14% versus prior best results on the 636-pair VinTAGeBench and achieved state-of-the-art VGGSound results.

Torching: PyTorch-Style ML Library from Scratch — KoldAd1b/torching

- Implemented reverse-mode autodiff and gradient accumulation over NumPy/CuPy-backed tensors, with CPU/GPU movement, broadcasting, and operator overloading.

Anvil: Self-Hostable Managed Agent Service — KoldAd1b/anvil

- Executes agent code in Docker/Modal sandboxes; models every run as a Temporal workflow over a Postgres event log, so runs survive process restarts.
- Streams live run output to a React client over NATS/SSE, with pgvector retrieval, GitHub App triggers, and scheduled runs; built in Bun/TypeScript with Hono.

Non-Elastic Container: Self-Hosted Platform-as-a-Service — KoldAd1b/non-elastic-container

- Deploys from a Git push, a prebuilt image, or a Compose file onto Docker Swarm, with zero-downtime rolling releases and automatic HTTPS through Caddy.
- Adds per-PR preview environments, multi-tenant workspaces, a browser terminal, and Loki/Grafana observability; orchestration in FastAPI and Temporal.

Publications

Accepted

- **A. Adib**, S. Ahmed, L. Khan. "Model Selection and KV-Cache Allocation Are Coupled under Memory Constraints." *NeurIPS 2026 LCFM Workshop*.
- **A. Adib**, S. Ahmed. "StallBound: The Memory-Bandwidth Limit of Chunked Prefill." *NeurIPS 2026 Workshop on On-Device Intelligence: Foundation Models under Real-World Constraints*.
- **A. Adib**, S. Ahmed, L. Khan. "FirmVLA: Task-Grounded Sequence-SLO Serving for Heterogeneous Robot Policies." *CoRL 2026 Workshop on Efficient Foundation Models for Real-Time Embodied AI*. Earlier version: "FirmVLA: Sequence-SLO Serving for Heterogeneous VLA Fleets." *NeurIPS 2026 Workshop on On-Device Intelligence: Foundation Models under Real-World Constraints*.
- **A. Adib**, S. Ahmed, L. Khan. "The Unseen-Mode Problem: Certifying Coverage Failures in Molecular Generators." *NeurIPS 2026 Workshop on Machine Learning for Simulations in Biology and Chemistry*.
- **A. Adib**, S. Ahmed, L. Khan. "Decomposition-Invariant Entitlement: Fair Scheduling for Agentic LLM Programs." *NeurIPS 2026 Workshop on AgenticOS: Co-designing Systems and ML Foundations of an OS Layer for Agentic AI*.
- **A. Adib**, S. Ahmed, I. Karim, L. Khan. "When Does Planning with a World Model Need Search?" *CoRL 2026 Workshop on Bringing Physics Simulation and World Models Together for Robotic Manipulation*.
- S. Ahmed, **A. Adib**, P. K. Saha, L. Khan. "What Do Expert Interventions Measure in Mixture-of-Experts Models?" *NeurIPS 2026 Workshop on Interpretability as a Science: Toward Rigorous Foundations for Understanding LLMs*.
- P. K. Saha, S. Ahmed, M. S. Zaoad, **A. Adib**, L. Khan. "Reading Where Distillation Writes: Interpretability-Derived Adapter Placement for Knowledge Distillation." *EACL*.
- "Quanta: A Fine-Grained Dataset and Benchmark for Arabic Machine-Translation Error Analysis." *SIGARAB*.
- **A. Adib**, M. H. Bukhari, U. Haque. "When Growing Food Is No Longer Enough: Indigenous Food Security Study." *Agriculture & Food Security*.

Under Review

- **A. Adib**, S. Ahmed, P. K. Saha, L. Khan. "Reliability-Gated Transport Tests for Mechanistic Interventions in Diffusion Models." *AAAI 2027*.
- S. Ahmed, P. K. Saha, **A. Adib**, M. S. Zaoad, L. Khan, P. Ranade. "Let it Settle: Allocating Test-Time Compute in Looped Language Models." *ICLR 2027*.